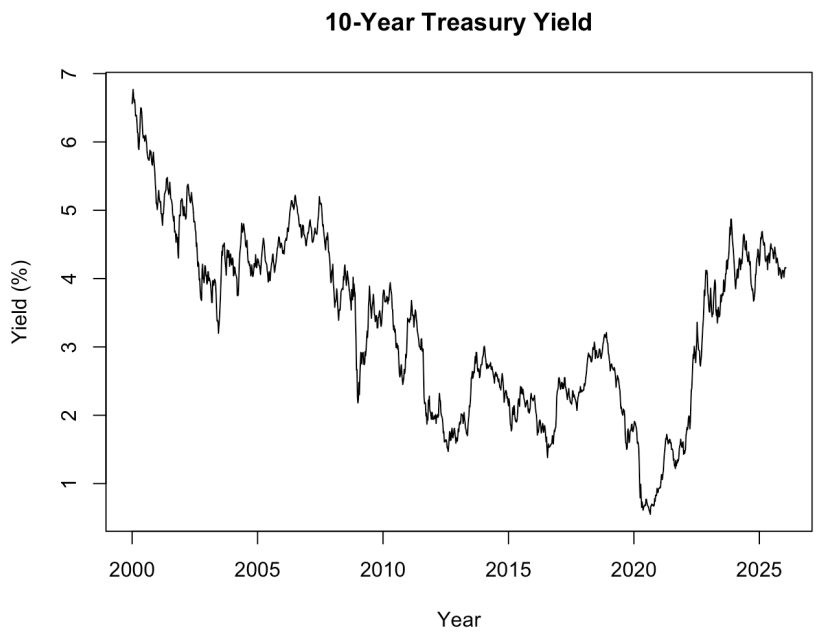
Kelly Valencia
March 25, 2026
Advanced Analytics

Treasury Yield Forecasting Analysis

Part 1:

1. The numerical value of R (the in-sample estimation period used to construct first forecast is): 574
   The numerical value of P (the out-of-sample forecast period is): 782
   The numerical value of T (the total sample size is): 1356

2.



3. First 10 Out-of-Sample Forecast Errors (h = 1)

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> print(errors_10)
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	Date	Model_1A	Model_1B	Model_1C	RWD	RW
1	2011-01-07	0.02	0.0625	0.1892308	0.025549738	0.02
2	2011-01-14	-0.04	-0.0325	0.1575000	-0.034494774	-0.04
3	2011-01-21	0.06	0.0425	0.2253846	0.065565217	0.06
4	2011-01-28	-0.02	0.0100	0.2100000	-0.014548611	-0.02
5	2011-02-04	0.14	0.1450	0.3550000	0.145476603	0.14
6	2011-02-11	0.14	0.2500	0.4973077	0.145224913	0.14
7	2011-02-18	-0.08	0.0900	0.4175000	-0.075025907	-0.08
8	2011-02-25	-0.14	-0.0950	0.2801923	-0.134896552	-0.14

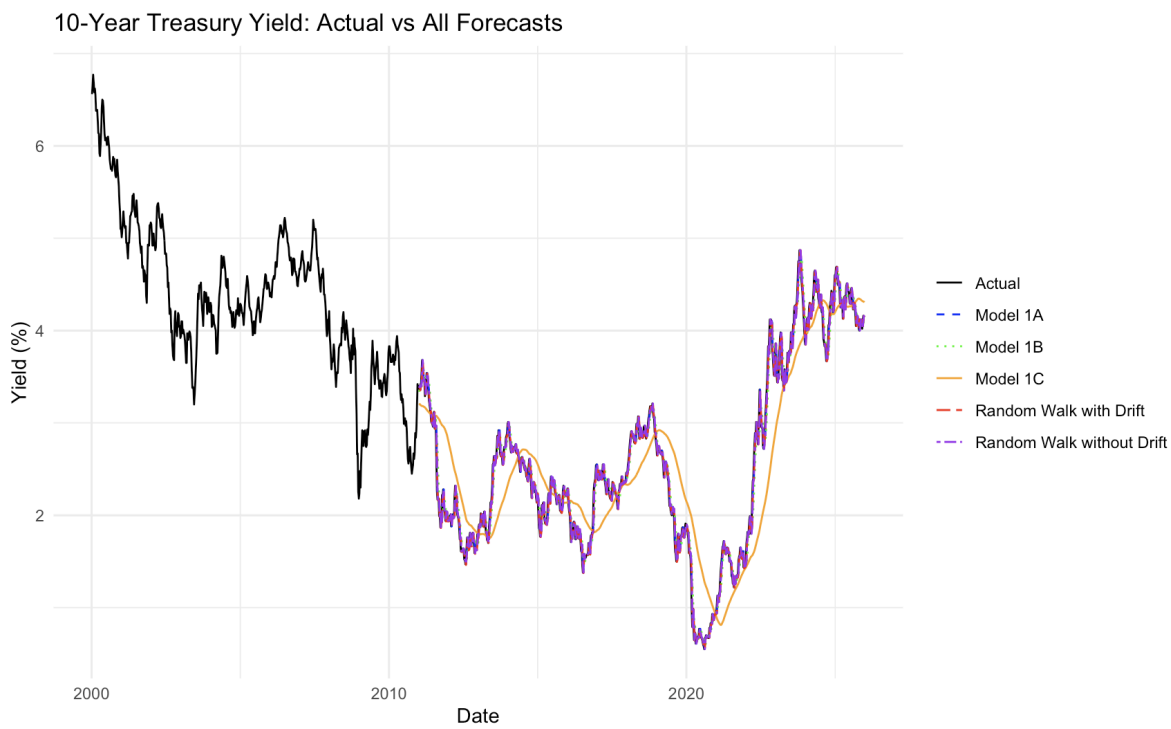
9	2011-03-04	0.01	-0.1000	0.2946154	0.015335628	0.01
10	2011-03-11	-0.01	-0.0925	0.2876923	-0.004690722	-0.01

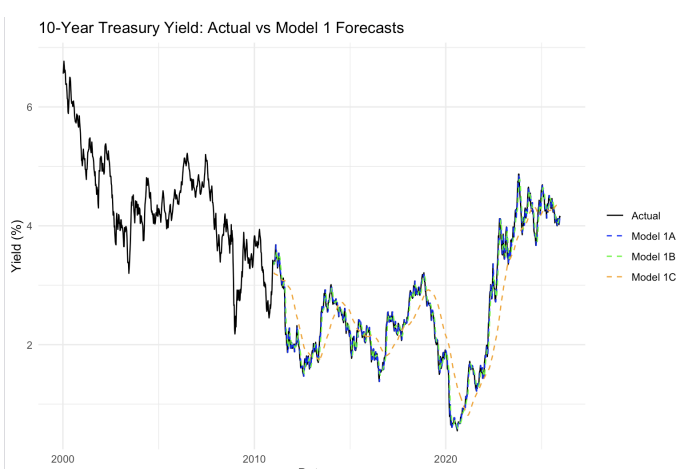
First 10 Out-of-Sample Forecasts (h = 1)

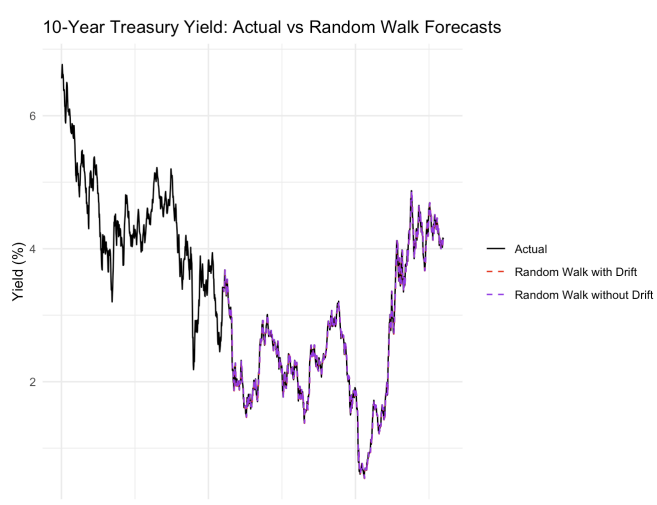
> print(forecasts_10)

	Date	Model_1A	Model_1B	Model_1C	RWD	RW
1	2011-01-07	3.38	3.3375	3.210769	3.374450	3.38
2	2011-01-14	3.40	3.3925	3.202500	3.394495	3.40
3	2011-01-21	3.36	3.3775	3.194615	3.354435	3.36
4	2011-01-28	3.42	3.3900	3.190000	3.414549	3.42
5	2011-02-04	3.40	3.3950	3.185000	3.394523	3.40
6	2011-02-11	3.54	3.4300	3.182692	3.534775	3.54
7	2011-02-18	3.68	3.5100	3.182500	3.675026	3.68
8	2011-02-25	3.60	3.5550	3.179808	3.594897	3.60
9	2011-03-04	3.46	3.5700	3.175385	3.454664	3.46
10	2011-03-11	3.47	3.5525	3.172308	3.464691	3.47

4.







Model 1A, Model 1B, and both Random Walk models (with and without drift) closely track the actual data, with only minor deviations. In contrast, Model 1C appears more smoothed compared to the actual series, indicating that it does not capture the volatility of the data.

For clarity, I plotted the Model 1 forecasts and the Random Walk forecasts separately, since they are very similar to the actual data and can be difficult to compare when combined. When viewed separately, Model 1A follows the actual data more closely than Model 1B, which is consistent with the MSFE results. Specifically, Model 1A has an MSFE of 0.008, which is lower than Model 1B's MSFE of 0.0204. But, the model that appears to follow the actual series most closely would be the Random Walk models.

5.

	Model	MSFE	RMSE
1	Model 1A	0.008399488	0.09164872
2	Model 1B	0.020439082	0.14296532
3	Model 1C	0.248328736	0.49832593
4	Random Walk with Drift	0.008425649	0.09179133
5	Random Walk without Drift	0.008399488	0.09164872

6. The best performing model is the Random Walk without Drift, although it is essentially indistinguishable from the Random Walk with Drift since their MSFE and RMSE values are almost identical. The RW with drift has an MSFE of 0.00842, while the RW without drift has an MSFE of 0.00839, so the difference is very small. Both models clearly outperform the others. Model 1C performs the worst, with much higher MSFE and RMSE. This model uses the average of the past year to produce a one-step-ahead (1 week ahead) forecast, which does not capture the patterns of the data well.

Part 2:

IPt = Industrial production , gIPt = growth rate of industrial production
FFR = federal funds rate
i)
The model I would use would be:
$gIPt = \beta 0 + \beta 1 gIPt\text{-}1 + \beta 2 FFRt\text{-}1 + \delta 2 Febt + \ldots + \delta 12 Dect + et$

I decided to model industrial production in growth rates rather than levels because the growth rate is more likely to be stationary and helps avoid spurious results. For my model, I included lagged components of industrial production since past industrial growth can still affect the

current value. Firms don't immediately adjust production, so this captures persistence and factors such as existing inventories. I also included a lagged value of the federal funds rate. According to economic theory, higher interest rates reduce investment and aggregate output. In addition, changes in monetary policy don't take effect immediately, so including a lagged term allows the model to capture delayed effects on industrial production. Lastly, I included seasonal dummy variables, with January as the base month, since industrial production can vary across seasons. For example, production may be higher during certain periods such as the holiday season. More precisely, the lagged industrial production growth term measures how a one unit increase in last month's industrial production growth is associated with a $\beta 1$ change in current industrial production growth, holding other factors constant. The coefficient on the federal funds rate captures the impact of monetary policy. If the coefficient is negative, this implies that higher interest rates reduce industrial production growth. In particular, a one unit increase in last month's federal funds rate is associated with a $\beta 2$ change in current industrial production growth, holding other factors constant. The seasonal dummy coefficients measure how industrial production growth in each month differs relative to January. This model allows the Federal Reserve to analyze the impact of monetary policy by examining the signs and magnitudes of the federal funds rate coefficient. The model can also be used for forecasting because it expresses future industrial production growth as a function of variables that are known at time t, such as past growth, lagged interest rate, and seasonal patterns. For this reason, I include only a lagged value of the federal funds rate, since including the current rate would require knowledge of future interest rates when forecasting.

ii) First, I would transform industrial production into its growth rate to help make the series stationary. Then I would split the data into a training sample and a testing sample, where about 70-80% of the earlier observations are used for training and the remaining 20-30% of the most recent observations are used for testing. I would estimate the model on the training sample and use it to generate a one step ahead forecast of industrial production growth. After that, I would expand the sample by one month, re-estimate the model, and make the next one step ahead forecast. I would keep repeating this recursive process until I reach the end of the testing sample. As time moves forward, the lagged growth rate of industrial production and the lagged federal funds rate update automatically. The seasonal dummy variables also update depending on the month being forecasted. After estimating the model on the training sample, I would look at the residuals and their ACF to check if there are any patterns left that the model didn't capture. I would also look at the coefficients and their t statistics to see if they are statistically significant. Finally, I would also try a simpler model, like one without the seasonal dummy variables, and compare the forecasts using MSFE to see if the simpler model performs better out of sample.